

KEMEROVO NOVOSTROYKA, Russian Federation

WMO: 296450

Lat: 55.250N

Lon: 86.217E

Elev: 148

StdP: 99.56

Time Zone: 7.00 (E07)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -37.8	(c) -34.7	(d) -40.9	(e) 0.1	(f) -37.6	(g) -38.0	(h) 0.1	(i) -34.6	(j) 6.4	(k) -5.1	(l) 5.5	(m) -7.3	(n) 0.4	(o) 140	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	29.6	18.7	27.8	18.1	26.1	17.4	20.2	26.3	19.4	25.3	18.7	24.1	2.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.4	22.4	17.4	12.7	21.7	16.6	12.1	21.0	59.0	26.4	56.3	25.1	53.8	24.3	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.1	5.3	4.6	-39.6	32.6	3.5	1.5	-42.2	33.7	-44.2	34.6	-46.2	35.5	-48.8	36.6
			-39.7	21.9	3.5	0.7	-42.3	22.4	-44.3	22.8	-46.3	23.2	-48.8	23.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.5	-19.1	-16.1	-5.4	4.5	9.8	17.3	18.9	16.5	10.0	2.7	-7.9	-14.7		
	DBStd	14.50	9.02	8.22	5.79	4.97	4.94	4.17	3.10	3.10	4.12	5.23	8.02	9.48		
	HDD10.0	3943	902	731	478	176	67	4	0	1	51	231	538	765		
	HDD18.3	6263	1161	964	737	416	268	66	32	74	251	483	788	1023		
	CDD10.0	828	0	0	0	9	60	224	275	202	51	6	0	0		
	CDD18.3	105	0	0	0	0	2	36	48	17	1	0	0	0		
	CDH23.3	1144	0	0	0	4	50	454	442	168	26	0	0	0		
	CDH26.7	279	0	0	0	1	5	123	117	28	5	0	0	0		
(14) Wind	WSAvg	2.1	1.9	1.9	2.2	2.4	2.7	2.0	1.7	1.7	1.9	2.1	2.4	2.2		
(15) Precipitation	PrecAvg	557	30	25	26	28	43	68	76	68	49	49	49	46		
	PrecMax	699	57	57	65	57	92	130	124	146	93	85	74	78		
	PrecMin	401	3	1	5	3	9	26	9	29	11	27	20	9		
	PrecStd	73	14	12	15	13	20	28	29	26	22	14	14	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.5	1.8	9.0	22.8	26.9	31.9	31.8	29.6	27.0	19.5	9.5	2.1		
		MCWB	-0.1	-0.1	4.1	12.4	14.8	18.6	19.7	19.7	16.1	11.9	6.5	0.8		
	2%	DB	-0.9	-0.2	5.9	18.7	24.8	29.7	29.6	27.2	22.4	15.8	5.6	0.9		
		MCWB	-2.0	-1.6	2.3	10.0	13.2	18.4	19.0	19.0	14.9	10.1	3.2	-0.2		
	5%	DB	-3.2	-1.4	4.1	15.9	22.4	28.0	27.8	25.2	19.8	12.9	3.1	-0.4		
		MCWB	-4.1	-2.5	1.3	8.6	12.5	17.7	18.8	17.7	13.4	8.6	1.4	-1.4		
	10%	DB	-5.6	-4.0	2.7	13.0	19.2	26.1	25.9	23.3	17.6	10.0	1.6	-2.5		
		MCWB	-6.4	-5.0	0.5	7.4	11.5	17.0	18.1	17.0	12.4	7.0	0.4	-3.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	0.1	4.6	12.7	16.4	21.2	21.5	21.0	17.3	12.6	6.5	1.0		
		MCDB	1.1	1.2	8.0	20.3	23.3	27.4	28.0	27.0	23.9	17.7	8.4	1.7		
	2%	WB	-2.0	-1.4	2.8	11.2	14.7	19.7	20.4	19.6	15.6	10.8	3.7	0.0		
		MCDB	-1.2	-0.3	5.0	17.6	21.7	26.5	26.6	24.9	21.2	15.1	5.5	0.7		
	5%	WB	-4.1	-2.6	1.7	9.5	13.5	18.9	19.7	18.7	14.3	8.9	1.6	-1.4		
		MCDB	-3.2	-1.3	3.7	15.1	20.3	25.7	25.6	23.6	18.9	12.7	2.8	-0.4		
	10%	WB	-6.5	-5.0	0.5	7.5	11.9	17.9	19.0	17.8	12.9	7.2	0.5	-3.6		
		MCDB	-5.7	-4.0	2.6	12.5	18.7	24.0	24.3	22.2	16.4	9.8	1.6	-2.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	9.8	9.7	10.3	11.2	12.0	10.4	10.5	10.7	7.7	6.7	7.6		
		MCDBR	7.9	8.8	10.2	16.0	16.1	15.8	14.4	13.8	16.1	13.8	6.0	7.2		
	5% WB	MCWBR	7.4	8.1	7.5	8.7	7.6	6.3	5.6	6.4	8.6	8.2	4.5	6.6		
		MCWBR	7.6	8.6	8.7	14.4	13.5	13.0	11.7	11.6	13.6	13.0	5.8	6.7		
(40) Clear-Sky Solar Irradiance	taub		0.297	0.333	0.382	0.409	0.413	0.411	0.437	0.414	0.378	0.362	0.333	0.303		
	taud		1.997	1.949	1.888	2.022	2.190	2.288	2.216	2.261	2.304	2.207	2.015	1.946		
	Ebn at Noon		575	695	752	802	828	835	802	796	774	673	517	454		
	Edh at Noon		71	109	147	149	134	123	130	117	98	84	70	60		
(44) All-Sky Solar Radiation	RadAvg		0.88	1.88	3.22	4.38	5.03	5.88	5.32	4.36	2.89	1.46	0.73	0.54		
	RadStd		0.10	0.14	0.31	0.28	0.59	0.51	0.38	0.34	0.36	0.17	0.13	0.09		

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page